

Homework 4

Due at 11:59 pm, May 11th

1 Linear Programming

Use the KKT conditions to solve the following LP problem:

$$\begin{aligned} \min_{x_1, x_2} \quad & x_1 + x_2 \\ \text{subject to} \quad & x_1 \geq -2, \\ & x_1 + 2x_2 \geq -2. \end{aligned}$$

Please clearly show the following steps:

- (a) Rewrite the constraints in the standard form

$$g_i(x) \leq 0.$$

- (b) Write down the Lagrangian function.

- (c) Write down all KKT conditions, including primal feasibility, stationarity, dual feasibility, and complementary slackness.

- (d) Use the KKT conditions to solve the problem.

- (e) Briefly explain which constraint or constraints are active at the optimal solution.

2 Quadratic Programming

Consider the following QP problem:

$$\begin{aligned} \min_{x_1, x_2} \quad & \frac{1}{2}(x_1^2 + x_2^2) - 2x_1 - x_2 \\ \text{subject to} \quad & x_1 + x_2 \leq 1, \\ & x_1 \geq 0, \\ & x_2 \geq 0. \end{aligned}$$

Please answer the following questions:

- (a) First ignore the constraints and find the unconstrained minimizer. Is this point feasible?

- (b) Rewrite the constraints in the standard form

$$g_i(x) \leq 0, \quad i = 1, 2, 3.$$

- (c) Write down the Lagrangian function.
- (d) Write down all KKT conditions, including primal feasibility, stationarity, dual feasibility, and complementary slackness.
- (e) Use the KKT conditions to solve the QP problem.
- (f) Briefly explain geometrically why the constrained minimizer is different from the unconstrained minimizer.

3 Dido's Problem



Figure 1: Dido's problem.

Here is a modified version of Dido's problem, named after Dido, the legendary founder and first queen of Carthage. According to the legend, Dido was given a bull's hide and told she could take as much land for her new city as she could enclose. She cut it into one long strip of leather, which was then used to form a semicircle whose diameter formed the coastline.

So they reached the place where you will now behold mighty walls and the rising towers
of the new town of Carthage;
and they bought a plot of ground named Byrsa ...
for they were to have as much as they could enclose within a bull's hide.

Virgil, *Aeneid*

Now instead of forming a circle or semicircle, let us limit ourselves to a rectangle. Specifically, assume a rope of length $l > 0$. We would like to find a rectangle that has the maximum area subject to the constraint that its perimeter does not exceed the length of the rope.

Let x_1 and x_2 denote the lengths of the two adjacent sides of the rectangle.

- (a) Formulate this as a constrained minimization problem using x_1 and x_2 as decision variables.
- (b) Clearly define the objective function and all constraints. Please remember to include the nonnegativity constraints on the side lengths.
- (c) Write down the Lagrangian function.
- (d) Write down all KKT conditions, including primal feasibility, stationarity, dual feasibility, and complementary slackness.
- (e) Use the KKT conditions to solve the problem.
- (f) What is the shape of the optimal rectangle? What is the maximum area?